

Boyce-Codd Normal Form (BCNF)

Strict Superkey Rule, 3NF vs BCNF Comparison, Exam Traps & Decision Guide

Boyce-Codd Normal Form (BCNF) is a stricter version of Third Normal Form (3NF). It is frequently tested in GATE, UGC NET, university exams, and technical interviews as the standard for removing all functional dependency-based anomalies.

Exam-Ready Definition ★: A relation is in BCNF if, for every non-trivial functional dependency $X \rightarrow Y$, X is a Superkey of the relation.

1. 3NF vs BCNF Comparison

In 3NF, a functional dependency $X \rightarrow A$ is allowed if X is a superkey **OR** A is a prime attribute. BCNF completely removes this "RHS is Prime" relaxation option.

Normal Form	Main Condition for Non-Trivial FD ($X \rightarrow Y$)
1NF	Atomic values; no multivalued or composite attributes.
2NF	1NF + No partial dependency (<i>ext{Proper subset of CK} \rightarrow ext{Non-Prime}</i>).
3NF	2NF + X <i>ext{ is Superkey}</i> OR Y <i>ext{ is Prime Attribute}</i> .
BCNF	3NF + X <i>ext{ MUST be a Superkey}</i> (No exceptions!).

Golden Rule ★: In BCNF, the Left-Hand Side (LHS) of *every* non-trivial functional dependency must be a Superkey!

2. BCNF Satisfied Example

Consider relation STUDENT(Roll_No, Name, Voter_ID, Age)

Candidate keys are: **Roll_No** and **Voter_ID**.

Functional Dependencies:

Functional Dependency	LHS Attribute (X)	Is LHS a Superkey?	BCNF Status
$Roll_No \rightarrow Name$	$Roll_No$	YES (Candidate Key)	Passes ✓
$Roll_No \rightarrow Voter_ID$	$Roll_No$	YES (Candidate Key)	Passes ✓
$Voter_ID \rightarrow Age$	$Voter_ID$	YES (Candidate Key)	Passes ✓
$Voter_ID \rightarrow Roll_No$	$Voter_ID$	YES (Candidate Key)	Passes ✓

Since the determinant (LHS) of every FD is a candidate key / superkey, the table is in **BCNF**.

3. How to Check BCNF in Questions

Step 1: Find ALL Candidate Keys of relation R .



Step 2: List all non-trivial Functional Dependencies ($X \rightarrow Y$).



Step 3: Calculate attribute closure X^+ for the LHS of every FD.



Is X^+ containing all attributes of R ?

- YES: X is a Superkey $\Rightarrow$ **Passes BCNF** ✓
- NO: X is NOT a Superkey $\Rightarrow$ **Violates BCNF** ✗

4. Classic BCNF Violation Example

Consider relation $R(A, B, C)$ with FDs:

- $AB \rightarrow C$
- $C \rightarrow B$

Step 1: Candidate Keys

- $(AB)^+ = \{A, B, C\} \Rightarrow AB$ is a Candidate Key.
- $(AC)^+ = \{A, C, B\} \Rightarrow AC$ is a Candidate Key.

Candidate Keys: $\{AB, AC\}$ | Prime attributes: $\{A, B, C\}$

Step 2: Testing FDs for BCNF

FD	Is LHS a Superkey?	3NF Status	BCNF Status
$AB \rightarrow C$	YES (AB is Candidate Key)	Passes 3NF ✓	Passes BCNF ✓
$C \rightarrow B$	NO ($C^+ = \{C, B\} \neq R$)	Passes 3NF ✓ (RHS B is Prime)	Fails BCNF ✗ (LHS not superkey)

Result: Relation R is in **3NF**, but **NOT in BCNF** due to $C \rightarrow B$.

5. Why BCNF is Stricter than 3NF

Consider a functional dependency like $D \rightarrow A$ where D is not a superkey, but A is a prime attribute:

- In **3NF**: $D \rightarrow A$ is **ALLOWED** ✓ because condition 2 (RHS is Prime) is satisfied.
- In **BCNF**: $D \rightarrow A$ is **FORBIDDEN** ✗ because condition 1 (LHS is Superkey) fails.

6. Hierarchy & Important Exam Traps ⚠

$BCNF \implies 3NF \implies 2NF \implies 1NF$

- Every relation in BCNF is guaranteed to be in 3NF, 2NF, and 1NF.

• The reverse is NOT guaranteed:

- $2NF \rightarrow 3NF$ (A 2NF table may contain transitive dependencies).
- $3NF \rightarrow BCNF$ (A 3NF table may contain non-superkey determinants with prime RHS).

7. Memory Tricks & Summary Table

Normal Form	Keyword Memory Trick	Violation Rule
1NF	Atomic	Multivalued attributes or nested tables in a cell.
2NF	No Partial	Proper subset of Candidate Key $\rightarrow$ Non-Prime attribute.
3NF	Superkey OR Prime	LHS is non-Superkey AND RHS is non-Prime.
BCNF	Superkey ONLY	LHS is non-Superkey (Regardless of RHS prime status).

One-Line Revision:

1NF = Atomic $\rightarrow$ 2NF = No Partial $\rightarrow$ 3NF = Superkey OR Prime $\rightarrow$ BCNF = Superkey ONLY